



Lecture 15

In the previous lecture, we found two complex poles of a second order system. Now we would like to find the impulse response of this second order system.

In order to do so, we would like to rearrange the transfer function in a way so that we can apply the inverse Laplace transform to get the time

domain response.

Below are some of the standard Laplace transform and their inverse Laplace transform.

$$F(s) = \frac{s+a}{(s+a)^2 + b^2}$$

$$\downarrow \mathcal{L}^{-1}\{\}$$

$$f(t) = e^{-at} \cos bt$$

$$F(s) = \frac{b}{(s+a)^2 + b^2}$$

$$\downarrow \mathcal{L}^{-1}\{\}$$

$$f(t) = e^{-at} \sin bt$$

$$F(s) = \frac{a^v + b^v}{(s+a)^v + b^v}$$

$$\downarrow v^{-1}\{\}$$

$$f(t) = 1 - e^{-at} \left(\cos bt + \frac{a}{b} \sin bt \right)$$

The two roots of the denominator
are $s = -\sigma \pm j\omega_d$

$$s_1 = -\sigma + j\omega_d$$

$$s_2 = -\sigma - j\omega_d$$

Complex poles always come in
complex conjugate pairs.

$$H(s) = \frac{\omega_n^v}{s^v + 2\zeta\omega_n s + \omega_n^v}$$

$$\frac{Y(s)}{U(s)} = \frac{\omega_n^v}{s^v + 2\zeta\omega_n s + \omega_n^v}$$

$$= \frac{\omega_n^v}{(s - s_1)(s - s_2)}$$

$$= \frac{\omega_n^v}{\{s - (-\sigma + j\omega_d)\} \{s - (-\sigma - j\omega_d)\}}$$

$$= \frac{\omega_n^v}{\underbrace{(s + \sigma)}_s \underbrace{+ j\omega_d}_a \underbrace{(s + \sigma)}_s \underbrace{- j\omega_d}_a}$$

$$= \frac{\omega_n^v}{(s + \sigma)^v + \omega_d^v}$$

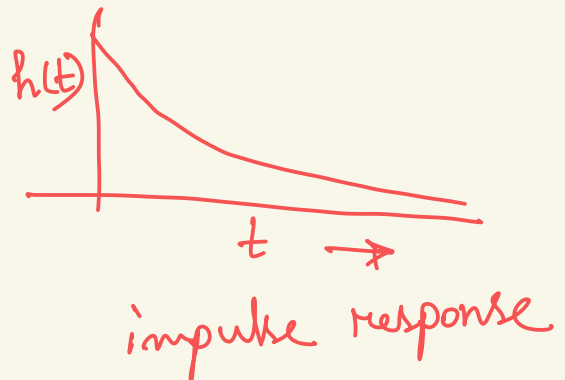
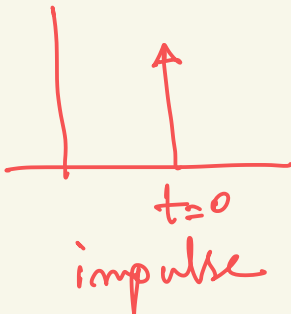
$$= \frac{\omega_n}{\sqrt{1-\zeta^2}} \cdot \frac{\omega_n \sqrt{1-\zeta^2}}{(s+\sigma)^2 + \omega_d^2}$$

$$H(s) = \frac{\omega_n}{\sqrt{1-\zeta^2}} \cdot \frac{\omega_d}{(s+\sigma)^2 + \omega_d^2}$$

By taking inverse Laplace transform we get the impulse response.

$$h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\sigma t} \sin \omega_d t, t \geq 0$$

$$\Rightarrow h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n t} \sin \omega_d t, t \geq 0$$



$h(t)$ is the impulse response.